Biology Extended Essay planner

Write your school's deadlines next to each stage. The online version is at whatthecell.org/extended-essay.html

What it is	An independent research essay of up to 4,000 words on a focused question, here in Biology
Time	About 40 hours of your own work, from November to June of Year 1, with the viva voce in November of Year 2
Marks	5 criteria, 30 marks in total, marked by an IB examiner
Reflection	Three formal reflection sessions with your supervisor, then a reflective statement of up to 500 words
Why it matters	Together with TOK it can add up to 3 points to your Diploma, and you need at least a D to pass the Diploma

These facts follow the IB *Extended essay guide* for essays first assessed in May 2027. Your school sets the exact deadlines, so copy them into the planner below.

> **EE or IA?** Your IA answers one focused question in 3,000 words with about 10 hours of work. Your EE goes deeper: 4,000 words, about 40 hours, a review of the published literature, and a fuller discussion of what your findings mean. The same work cannot count as both.

Two pathways

- **Subject-focused (Biology).** Your essay sits in Biology and uses biological methods and ideas. Most Biology students choose this.
- **Interdisciplinary.** Your essay combines Biology with a second DP subject, for example Biology with Psychology or Chemistry, and sits within one of five broad frameworks (such as science, technology and society). ESS cannot be combined with another subject: an essay mixing biology and geography should be written in ESS instead.

The five criteria in plain words

CRITERION	MARKS	LOOKED AT THROUGH	WHAT THE EXAMINER WANTS TO SEE
A. Framework for the essay	6	Research question, research methods, structure	A focused, analytical question; methods that suit it, described well enough to repeat; a clear structure with labelled figures
B. Knowledge and understanding	6	Knowledge, terminology, concepts	Accurate biology drawn from the literature and your own data, in correct scientific language
C. Analysis and line of argument	6	Analysis, line of argument	Findings broken down and backed by evidence, with a clear thread from question to conclusion

CRITERION	MARKS	LOOKED AT THROUGH	WHAT THE EXAMINER WANTS TO SEE
D. Discussion and evaluation	8	Discussion, evaluation	What the findings mean, how strong they are and what limits them, weighed throughout the essay
E. Reflection	4	Evaluative, growth	How the EE changed you as a learner, with real examples of skills you now use elsewhere

Criterion D carries the most marks, and evaluation should run **through** the essay, not sit in one section at the end. Most weak EEs stop at describing results.

The staged plan

Our timeline: you start in **November of Year 1**, hand in the final essay **before the summer holidays at the end of June in Year 1**, and have your viva voce in **November of Year 2**. Your supervisor will give you the exact deadlines, so copy them into the planner.

STAGE	WHEN	WHAT YOU DO	DONE WHEN
1. Choose a subject and supervisor	Year 1, November	Decide on Biology (subject-focused) or an interdisciplinary pathway. Read two or three strong Biology EEs. Set up your researcher's reflection space (a notebook or digital log)	You have a supervisor and a reflection space
2. Explore a topic	Year 1, November to December	Read widely around two or three topics. Note what is known, what is argued about and what you could measure or find data on	You can explain each topic in two minutes
3. Research question and first reflection session	Year 1, December	Narrow to one focused question. Meet your supervisor for the **first formal reflection session**: your ideas and how you plan to research them	Your question is agreed
4. Plan, safety and ethics	Year 1, January	Write a research plan: approach, variables, method, risk assessment and ethics check. Start your literature review	Your supervisor signs off the plan
5. Pilot and data collection	Year 1, February to March	Run a small pilot, fix problems, then collect your full data (or extract your secondary data). Log everything	You have all your data
6. Analysis	Year 1, March to April	Process the data, draw the graphs, run suitable statistical tests and treat uncertainties	Every result is in a table or graph
7. Interim reflection session and full draft	Year 1, April to early May	The **interim reflection session**: show your progress and discuss problems. Then write the full draft	Full draft to your supervisor
8. Feedback and final version	Year 1, May to end of June	Your supervisor gives written comments on your **full draft** (the only draft they may comment on in writing). Revise, proofread, check references, layout and word count yourself	Final essay handed in before the summer holidays

STAGE	WHEN	WHAT YOU DO	DONE WHEN
9. Viva voce and reflective statement	Year 2, November	The **final reflection session** (viva voce): a short conversation about the finished essay. Then write your reflective statement of up to 500 words on the reflection and progress form	Statement handed in

You can also check in with your supervisor between the three formal sessions. Those check-ins are useful but are not recorded on the form.

Checklist

- ☐ My question is focused, analytical and answerable in 4,000 words.
- ☐ I can get enough data in the time I have, with equipment the school has (or from reliable published sources).
- ☐ My plan has been checked for safety, ethics and environmental impact.
- ☐ I keep a researcher's reflection space: dates, decisions, problems, sources and photos.
- ☐ Every source is recorded as I read it.
- ☐ I have used and declared any AI tools in line with school rules.
- ☐ My draft discusses and evaluates throughout, not only at the end.
- ☐ My final word count is 4,000 or fewer.
- ☐ My reflective statement is 500 words or fewer and written after the viva voce.

Choosing a Biology research question

A Biology EE needs biology at its heart: living things, their parts or their processes. A strong question:

- is specific: it names the organism (in italics, *Genus species*), the variables or system, and what is measured
- invites analysis, not description: "to what extent", "how significant" or "what is the effect of" work well
- asks one thing, not two joined by "and"
- does not have an obvious answer before you start
- is realistic with school equipment and in the time you have

TOO BROAD	BETTER
How does pollution affect plants?	What is the effect of copper sulfate concentration (0 to 100 mg dm^{-3}) on the growth rate of *Lemna minor* over 14 days?
Is exercise good for you?	To what extent does the duration of moderate exercise affect recovery heart rate in 16 to 18 year olds? (with consent)
Do antibiotics work?	What is the effect of garlic extract concentration on the zone of inhibition of *Escherichia coli* K-12?
What lives in mangroves?	How does distance from the shore affect the species diversity of crabs in a mangrove forest?
Why are coral reefs dying?	To what extent is sea surface temperature correlated with coral bleaching at Southeast Asian reef sites from 2000 to 2020? (database)

Primary or secondary data?

Both are allowed in Biology, and every EE must include a review of the published literature.

- **Primary data:** a laboratory experiment or fieldwork with a controlled independent variable, repeats and a standard method. Simple school experiments are just as valid as university research.
- **Secondary data:** data taken from published studies or databases such as GBIF, NCBI or long-term climate and population records. Choose your studies using clear criteria that you state, make sure they used comparable methods, and do your own analysis. An essay that only summarises what others found will not score well on criteria C and D.

Ethics and safety

- No experiments that cause pain, harm or distress to animals. Watching animals in their habitat without disturbing them is fine.
- Human participants give informed consent (parental consent if under 16), can withdraw at any time, and stay anonymous.
- Follow the IB experimentation guidelines and your school's rules for microorganisms, chemicals, dissection and fieldwork. Leave field sites as you found them.

Structure of a Biology EE

SECTION	ABOUT HOW MANY WORDS	PURPOSE
Title page and contents	(not counted)	Title, research question, subject, word count. No names: not yours, your supervisor's or your school's
Introduction	300	The question, why it matters to you and to biology, and how you will investigate it
Background biology and literature review	700 to 900	The theory your discussion needs, and what published studies have already found
Method	500 to 700	Variables, materials, procedure, safety and ethics, in enough detail to repeat
Results and analysis	600 to 800	Processed data, graphs and statistics, each explained in words (tables and graphs do not count)
Discussion and evaluation	800 to 1,000	What the findings mean, compared with published work, with the strength and limits of the evidence
Conclusion	400	A clear answer to the question, its limits and what you would investigate next
References and appendices	(not counted)	Every source in one consistent style; raw data if needed

Format rules

- **Word count:** introduction, body, conclusion and quotations count. The contents page, headings, tables, graphs, diagrams, equations, calculations, citations and references do not. Examiners stop reading at 4,000 words.
- **Layout:** font size 12, 1.5 line spacing, page numbers from the page after the contents page, and a file under 10 MB.
- **Appendices and footnotes:** examiners do not have to read them, so never put anything important there.

- **No abstract** is needed.

What the top marks look like

- **A:** the question is precise and stays the same from title page to conclusion. The method could be repeated by someone else. Headings, numbered figures, units and references are consistent.
- **B:** the biology is accurate and goes beyond the textbook where needed, using journal articles and reliable sources, with correct terminology.
- **C:** every claim is backed by your data or sources, the statistics suit the data, and each section leads to the next.
- **D:** you explain what the results mean, admit what they cannot show, weigh how much each limitation matters, and compare with other studies, all through the essay.
- **E:** your statement shows how your thinking and skills changed, with specific moments, and how you now use those skills elsewhere.

The reflective statement

After the viva voce you write one reflective statement of up to 500 words on the reflection and progress form. It is the only evidence for criterion E, so a blank form scores zero.

Strong statements are personal and specific. They describe:

- how your question and approach changed, and why
- a setback and how you dealt with it
- what you would do differently next time
- the skills you gained and where you have used them since, in other subjects, CAS or plans for university

Your researcher's reflection space is the raw material: write in it after every session and whenever you make a decision.

Common mistakes

- A question so broad that the essay becomes a textbook summary.
- Starting data collection too late, especially with living organisms.
- Results with no statistics, or statistics that do not suit the data.
- A separate "evaluation" paragraph tacked on at the end instead of evaluation throughout.
- A conclusion that repeats the results instead of answering the question.
- Names of the student, supervisor or school left on the title page or in the acknowledgements.
- Writing the reflective statement from memory at the last minute.

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